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23 March 1978
SUPERSEDING
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17 November 1969

MILITARY SPECIFICATION

CORD, FIBROUS, POLYESTER, DIAMOND BRAID

This specification is approved for use by US Army Natick Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one size of diamond braided polyester cord.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.

MILITARY

MIL-C-3131 - Cordage; Preparation for Delivery of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4020

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.2 Government and contractor purchase. The requirements specified in 3.7 and 3.8 apply to cord purchased directly by the Government. All other requirements apply both to cord purchased by a contractor as a component for an end item and to cord purchased directly by the Government.

3.3 Standard sample. The cord shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.4 Material. The yarn used in the fabrication of the cord shall be a high tenacity, multi-filament polyester yarn.

3.5 Construction and physical requirements. The cord shall be of diamond braid construction without core, having eight strands braided over and under each other in a circular direction and conforming to the requirements specified herein. The stiffness of the finished cord shall be equal to the standard sample.

3.5.1 Diameter. The diameter of the finished cord shall be 3/16 inch maximum when tested as specified in 4.4.5.

3.5.2 Length per pound and breaking strength. The finished cord shall have a minimum of 105 feet per pound and have a minimum breaking strength of 1100 pounds when tested as specified in 4.4.5.

3.5.3 Elongation. The elongation of the finished cord shall not be more than 10 percent at a 825 pound load when tested as specified in 4.4.5.

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3.5.4 Picks per inch. The finished cord shall have 2.5 to 3 picks per inch when tested as specified in 4.4.5.

3.6 Color and finish. The color of the finished cord shall match the approved shade of Olive Drab 7 (see 6.3), when examined in accordance with 4.4.5. The coloration shall be imparted to the cord by a resin bonded pigment system applied from a solvent or solvent emulsion phase (see 4.4.2). The cord shall be uniformly impregnated with a synthetic vinyl finish (see 6.5). The percentage of color and finish based on the weight of the unfinished cord, shall be not less than 8 percent nor more than 15 percent when tested as specified in 4.4.5.

3.6.1 Colorfastness. The finished cord shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.4.5. When no standard sample has been established, the colored cord shall show "good" fastness to weathering when tested or specified in 4.4.5.

3.7 Put-up. Unless otherwise specified (see 6.2), the cord shall be furnished without knots, in continuous length 1000-foot spools (reels). Pieced lengths (lapped on ends) will be permitted but a spool (reel) shall contain no more than three pieces and no piece shall be less than 100 feet in length, or shall contain no more than two pieces with no minimum length applicable. The cord shall be wound on the spools (reels) so that each turn and layer is free from entanglement. The ends of the cord shall be served or heat sealed to prevent fraying. A plus tolerance of 10 percent shall be allowed on the length of spool (reel) specified.

3.8 Identification. Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and shall have a minimum tearing resistance in both directions (total) of 850 grams when tested as specified in UU-T-81. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. All ticket or label entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock number
Nomenclature
Specification number
Length
Contract number and date
Contractor's name

3.9 Workmanship. The finished braided cord shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.1 shall be inspected as specified in 4.4.3 and 4.4.4 for compliance with appearance, length, and winding. The sample shall be tested for physical and chemical characteristics in accordance with 4.4.5.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in the specification or applicable purchase documents. In addition, inspection shall be performed for the requirements in table I and contractor's certificate of compliance thereto shall be furnished and will be accepted for the requirement.

TABLE I. Component tests

Characteristic	Paragraph reference
Material identification	3.4
Finish identification	3.6

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* 4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether the operations are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be carried out to determine that coloration process (see 3.6) was carried out as specified. Unless otherwise specified, a certificate of compliance will be accepted for this requirement.

* 4.4.3 Examination of the end item for visual defects. The cord shall be examined by unwinding and the defects listed in table II shall be counted regardless of their proximity to each other. The sample unit for this examination shall be one spool (reel). The lot size for this examination shall be expressed in units of one spool (reel) each. Not less than 100 feet from each sample unit shall be subjected to the visual examination. The acceptable quality level shall be 1.5 major defects and 4.0 total defects (major and minor combined) per 100 units. The inspection level shall be level I.

TABLE II. Visual examination

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	X	
	Chafed or damaged	X	
	Kink	X	
	Knot	X	
	Not equal to standard sample for stiffness	X	
	Broken, projection, or loose end or strand	X	
Color	Not uniform		X
Cleanness	Spot or stain 1/		X
Identification ticket or label	Omitted, incorrect, illegible, insecurely attached		X
	Ticket or label not as specified		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

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* 4.4.4 Examination of the end item for length and winding. The sample unit for this examination shall be one spool (reel). The inspection level shall be level S-3 and the acceptable quality level (AQL) shall be 4.0 percent defective. The lot size shall be the number of spools (reels) in the lot. Defects shall be as listed in 4.4.4.1 and 4.4.4.2. Any spool (reel) found to contain one or more defects shall be classified as a defective unit. The length on the spool (see 3.7) shall be determined using the following formula:

W x L = Length per spool (reel) in feet

Where: W = net weight of the spool (reel) in 0.1 pounds
L = lot average of the weight-length relationship (length per pound) as determined in 4.4.5

4.4.4.1 Defects with regard to length. Defects shall be considered to exist if any of the following are determined during inspection:

(a) The total length is not as specified (including applicable tolerance).

(b) Length of cord is more than 6 feet less than length marked on ticket.

4.4.4.2 Defects with regard to winding. Defects shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding or otherwise affecting free unhampered unwinding of cord.

(b) Put-up not as specified i.e. not spools (reels).

(c) Any end not served or heat sealed.

(d) Any spool (reel) found to contain more than three pieces.

(e) Any piece less than 100 feet in length (except when there are only two pieces per spool).

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4.4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table IV, shall be followed. When the data in the "Number of determinations" or "Results reported as" columns are not specified in the table, it shall be as required by the referenced test method. The physical and chemical values specified in section 3, except as specified herein, apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. For the percentage of vinyl finish, the requirements shall apply only to the lot average (average of all sample units within the lot). The sample unit for the percent add on for the vinyl finish shall be 12 feet of unfinished cord and 12 feet of finished cord. One determination shall be made per sample unit. The sample size shall be five regardless of the lot size, and the results shall be averaged. The sample unit for end item testing shall be one spool (reel) of finished cord. The lot size shall be expressed in units of one spool (reel) of cord. The sample size shall be in accordance with table III. The lot shall be unacceptable if one or more units fail to meet any test requirement specified or if the lot average of the percentage of vinyl finish fails to meet the requirement specified in 3.6. All test reports shall contain the individual values utilized in expressing the final result. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, test shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE III. Sampling for tests

No. of units in lot	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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* TABLE IV. Test methods

Characteristic	Requirement reference	Test method	Number of determinations per sample unit	Results reported as
Construction (diamond braid)	3.5	Visual	1	Pass or fail
No. of strands	3.5	Visual	1	Pass or fail
Diameter	3.5.1	6002	-	-
Length per pound	3.5.2	4.4.5.1	-	-
Breaking strength	3.5.2	6016	-	-
Elongation	3.5.3	6016	-	-
Picks per inch	3.5.4	6001	-	Pass or fail
Color matching	3.6	4.4.5.2	1	Pass or fail
Percentage of finish	3.6	4.4.5.3	-	-
Colorfastness to weathering	3.6.1	5671 1/	-	-

1/ The time of exposure shall be 40 standard fading hours.

* 4.4.5.1 Determination of length per pound. A specimen of cord shall be put under a tension load of 11 pounds, and with the specimen still under load, a minimum length of 10 feet shall be accurately measured and marked on the specimen to the nearest 1/4 inch. After marking, the tension load shall be removed and length measured shall be cut out of the test specimen and accurately weighed to within plus or minus 1 percent of its total weight. Three determinations shall be made per sample unit and the results shall be averaged to the nearest 0.1 foot for the length per pound. Failure of any sample unit to meet specified requirement for length per pound constitutes failure of this test.

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* 4.4.5.2 Determination of color matching. The color of the finished cord shall match the standard sample under artificial daylight having a color temperature of 7500 kelvin and under incandescent lamplight at 2800 kelvin. Failure to match the color of the standard sample at 7500 kelvin or to closely approximate the standard sample at 2800 kelvin constitutes failure of the test.

* 4.4.5.3 Determination of percentage of color and finish. The percentage of color and vinyl finish impregnating material on the cord shall be determined by weighing an identical sample of cord, before and after the coloration impregnating process. The test specimen shall be a minimum of 10 feet of cord. Weight (length per pound) shall be determined in accordance with the applicable provisions of 4.4.5.1. The percentage of color and finish shall be calculated as follows:

$$\text{Percent finish} = \frac{B - A}{A} \times 100$$

Where: A = Length per pound of unfinished cord.
B = Length per pound of finished colored cord.

* 4.4.6 Packaging inspection. Preservation-packaging, packing, and marking shall be inspected in accordance with the quality assurance provisions of MIL-C-3131, except that the inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Cord shall be preserved-packaged in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Cord shall be packed in accordance with the applicable requirements of MIL-C-3131.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

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6. NOTES

6.1 Intended use. This cord is intended for use as a suspension line for the M-1966 jungle hammock.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.3, and 6.4).
(c) Put-up, if other than specified (see 3.7).
(d) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 Sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one spool (reel) of the finished cord. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Vinyl shade and finish. Information is furnished that the standard sample for shade and finish was developed by utilizing the Olsen Preservative and Paint Corporation (Whippany, NJ), synthetic vinyl finish SM-516.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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